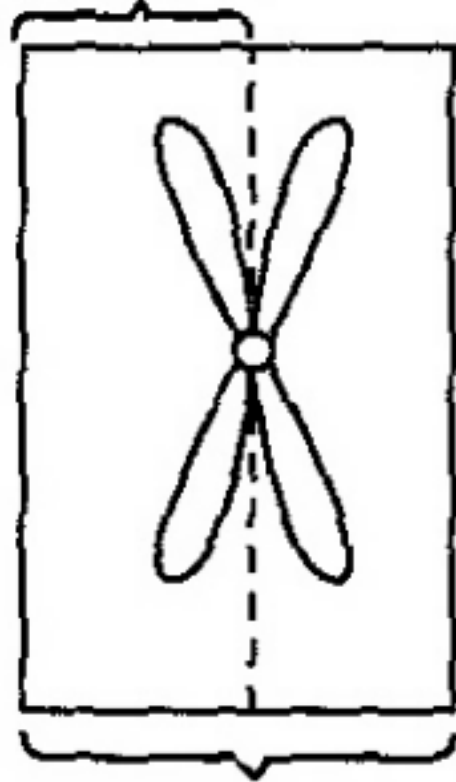


染色单体



染色体

(1)



(2)

各有几个**DNA**分子？

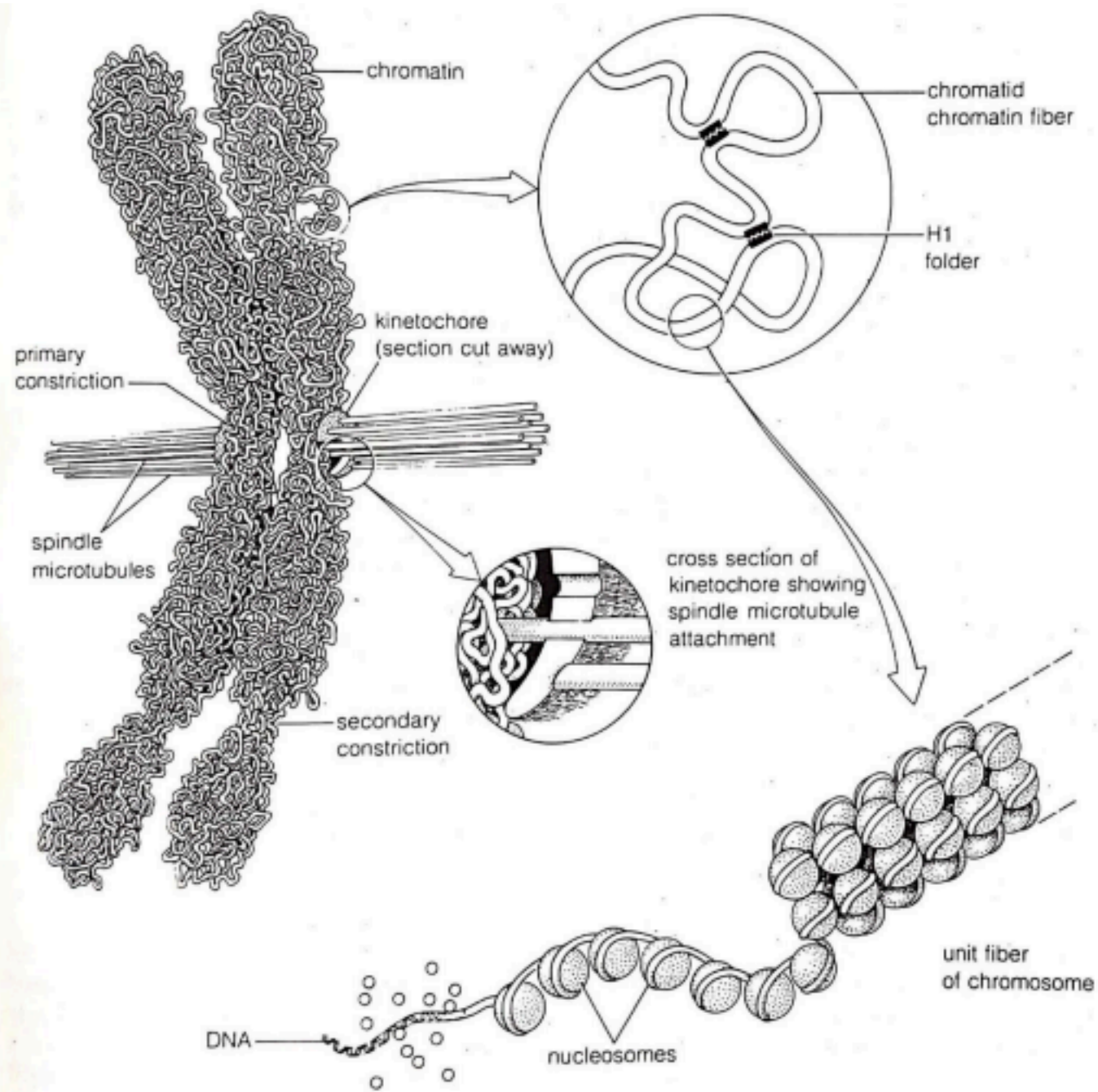


Figure 15-10 A summary of metaphase chromosome structure (see text).

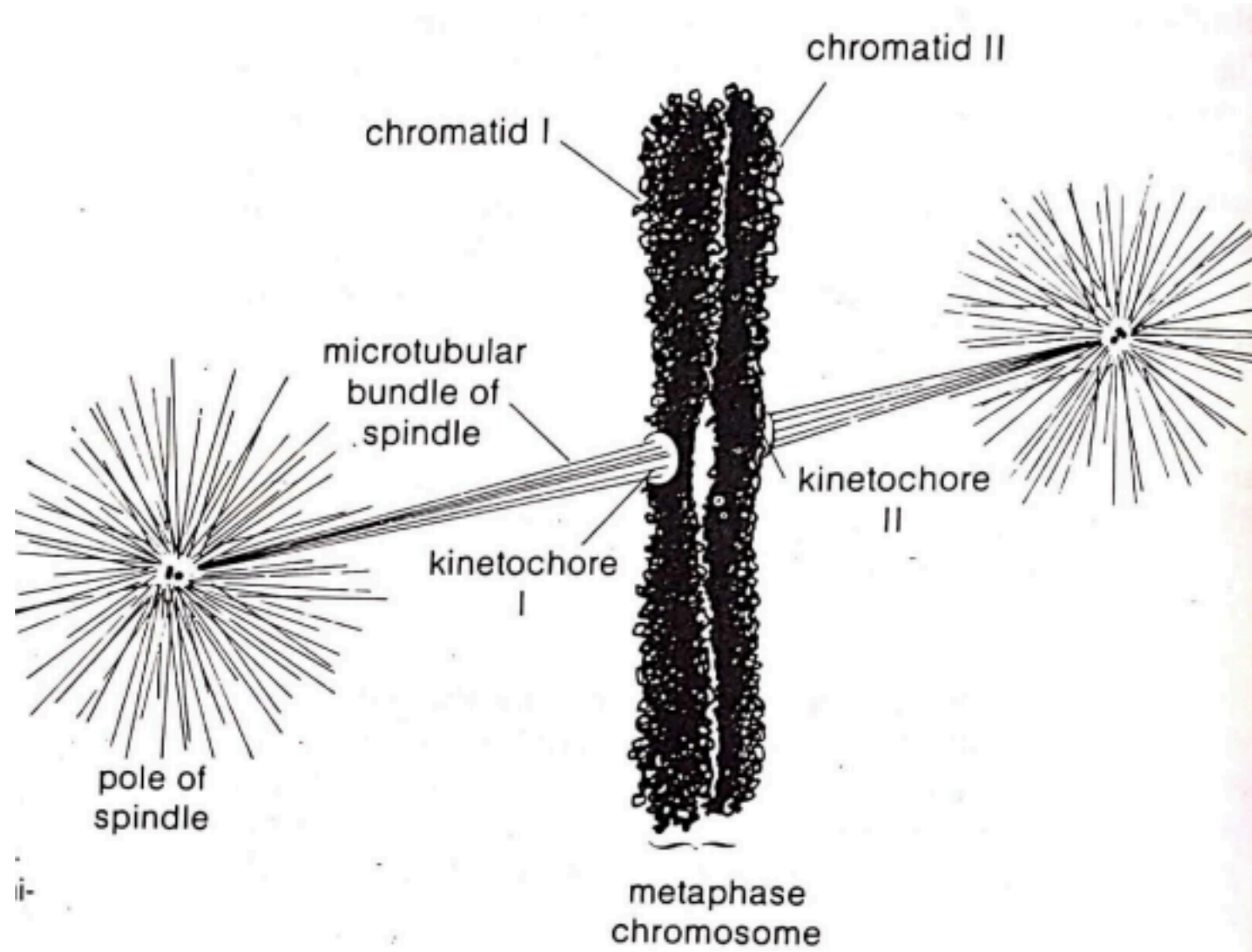


Figure 14–9 The orientation of kinetochores at mitotic metaphase. The two kinetochores of each metaphase chromosome connect to microtubules leading to opposite poles of the spindle.

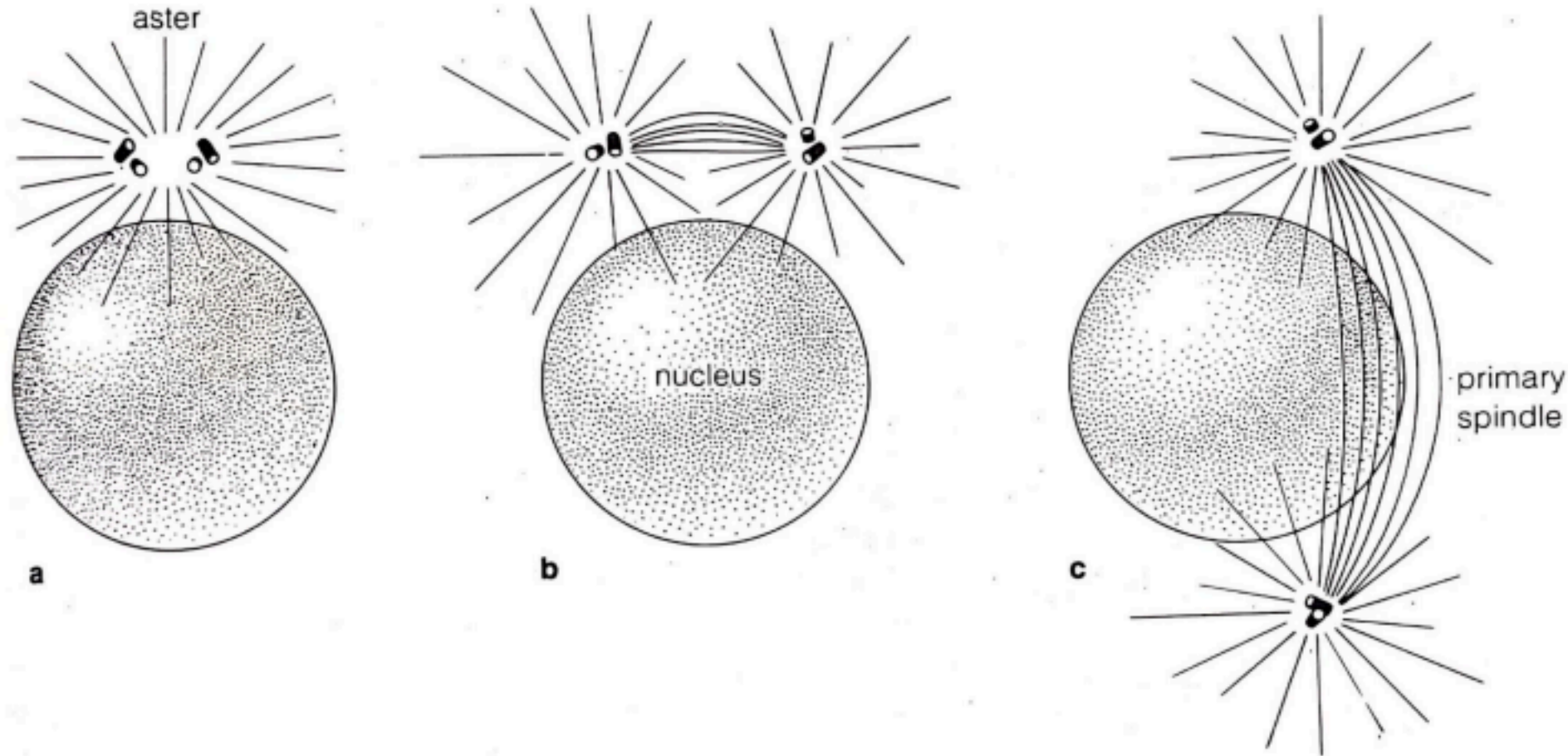
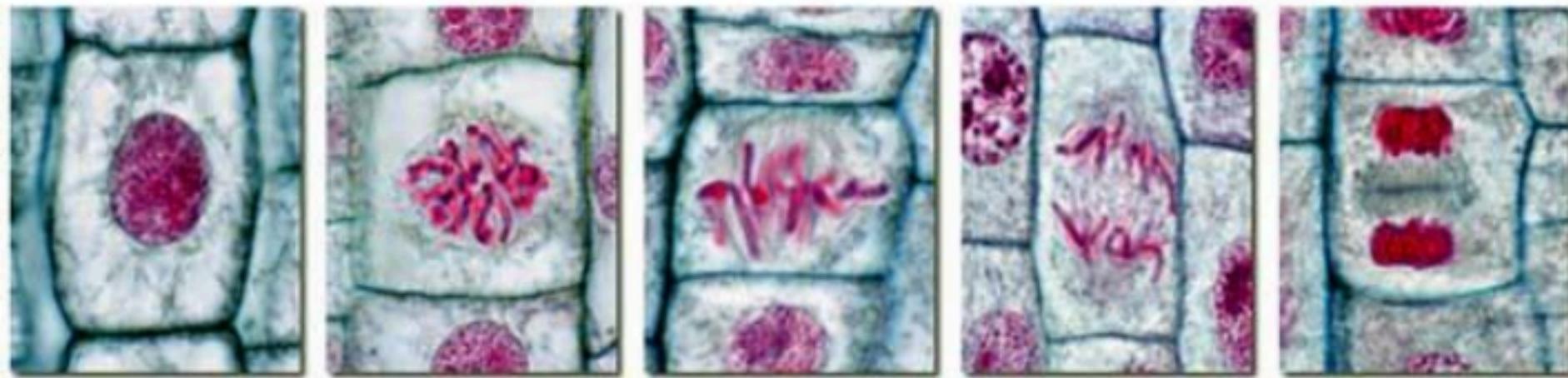
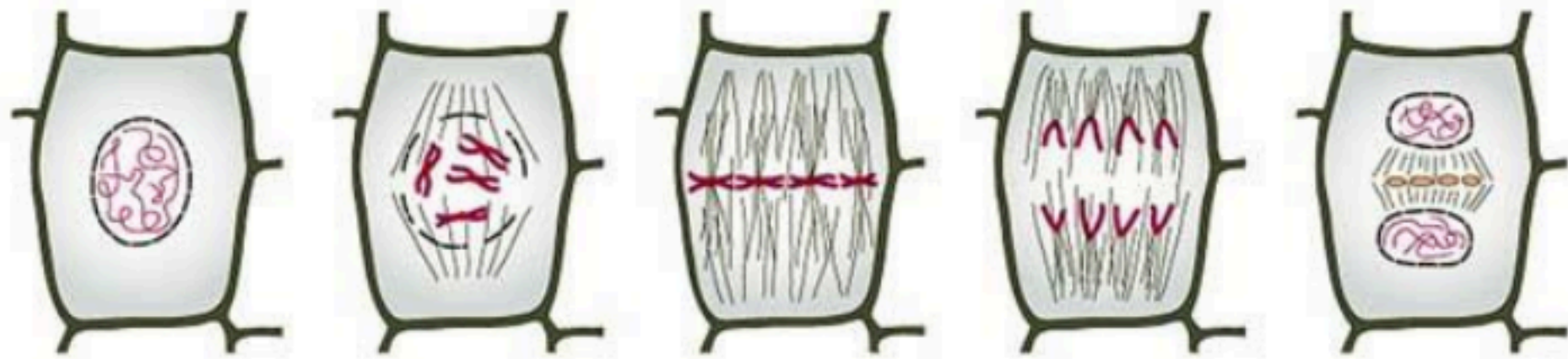
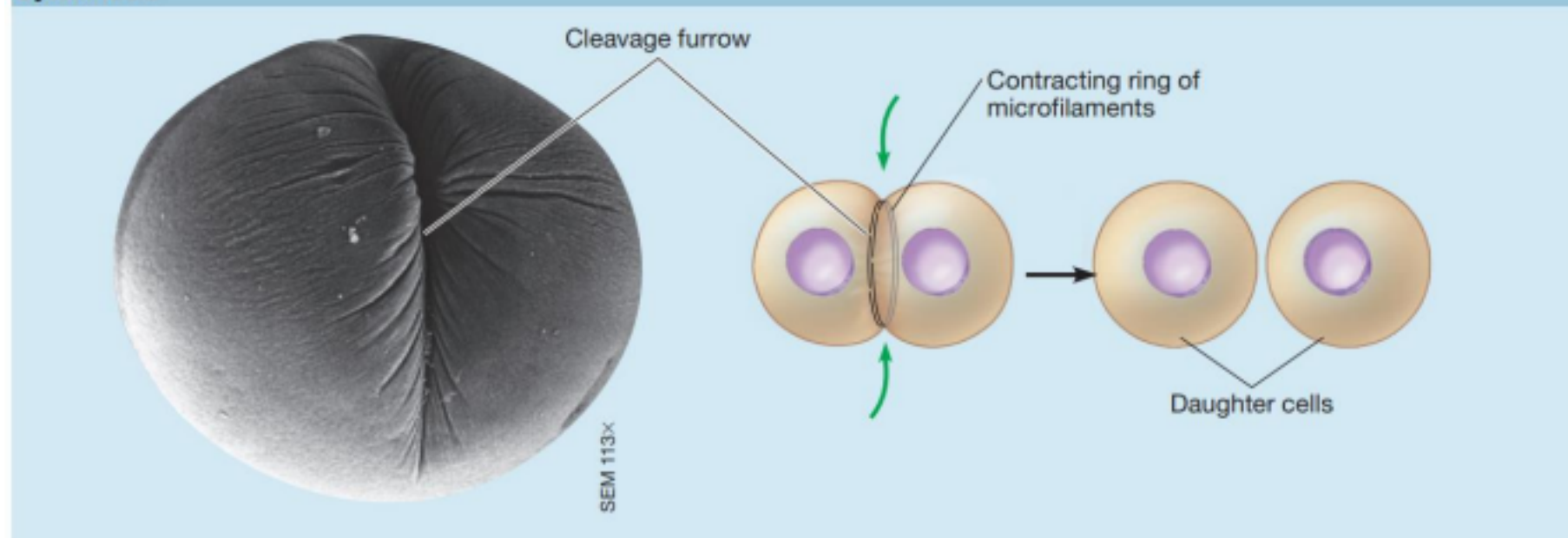


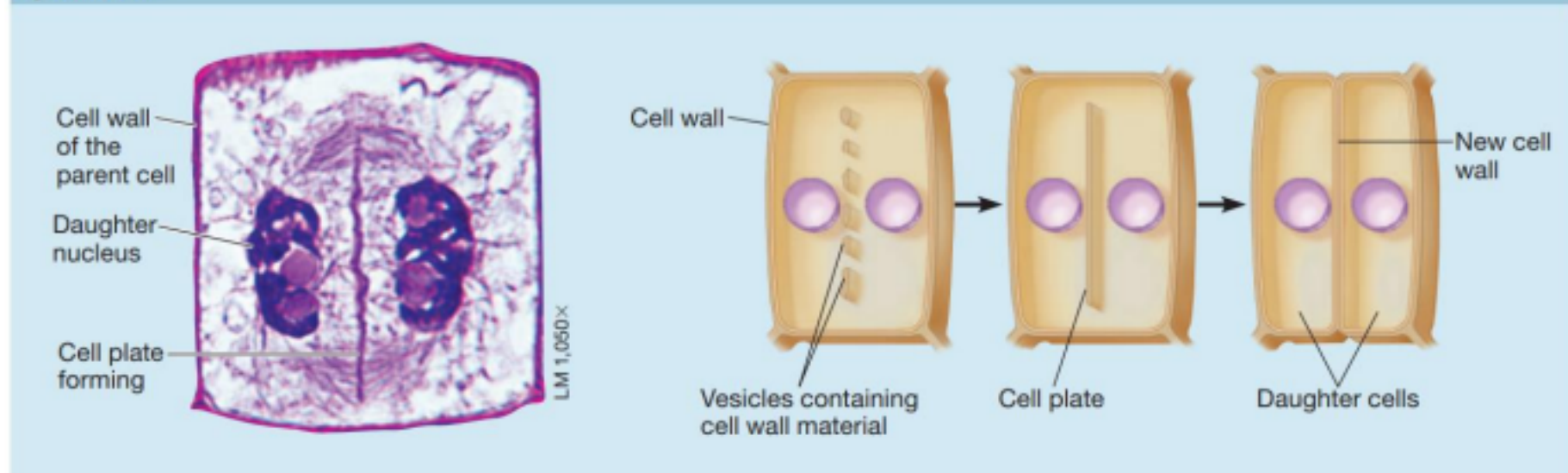
Figure 14–6 Movements of the centrioles and asters and formation of the primary spindle in prophase of mitosis (see text).



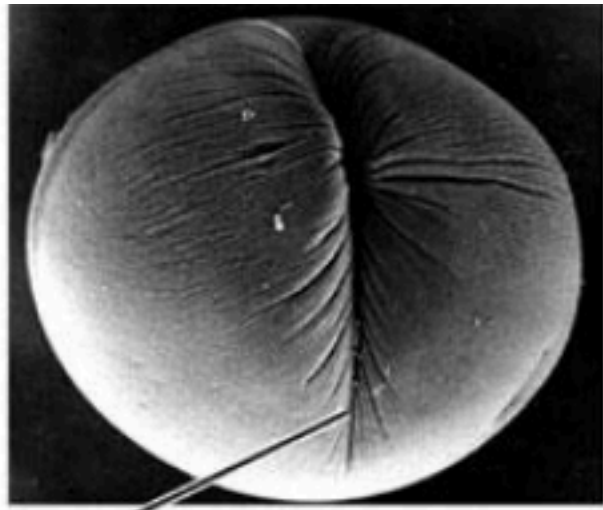
Cytokinesis



▲ **Figure 8.6A** Cleavage of an animal cell

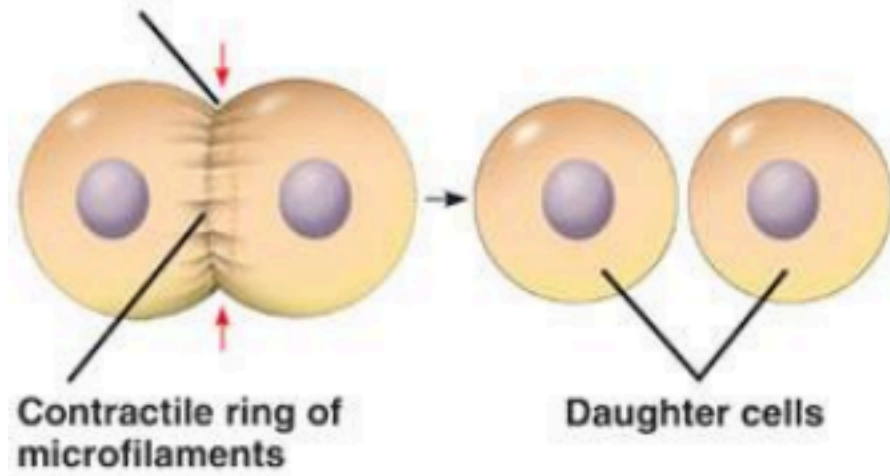


▲ **Figure 8.6B** Cell plate formation in a plant cell

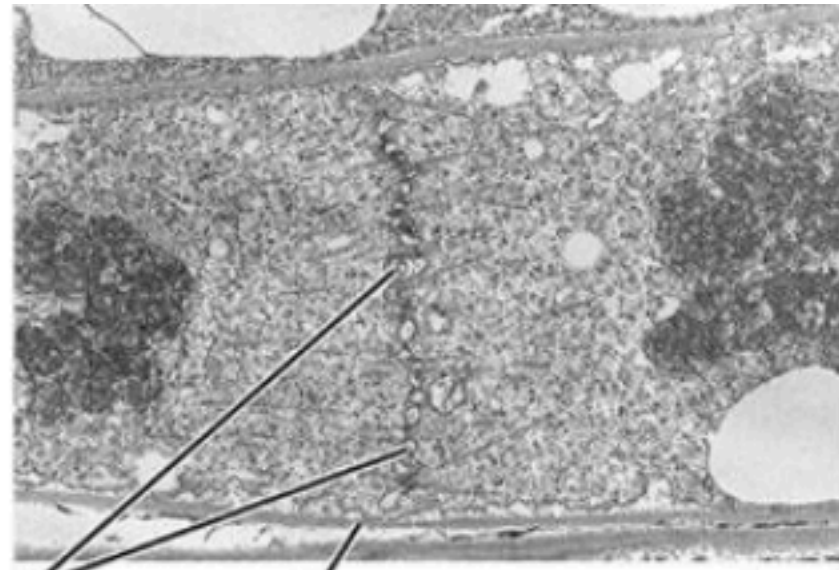


100 μm

Cleavage furrow



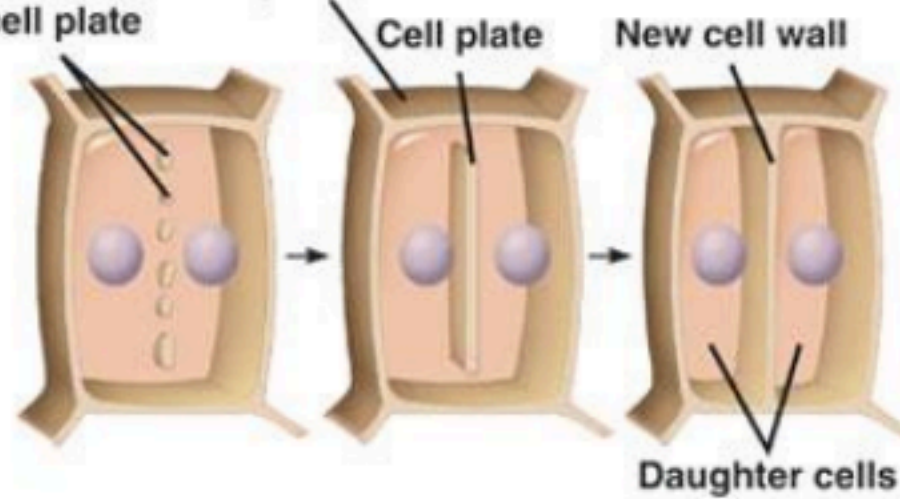
(a) Cleavage of an animal cell (SEM)



1 μm

Vesicles forming cell plate

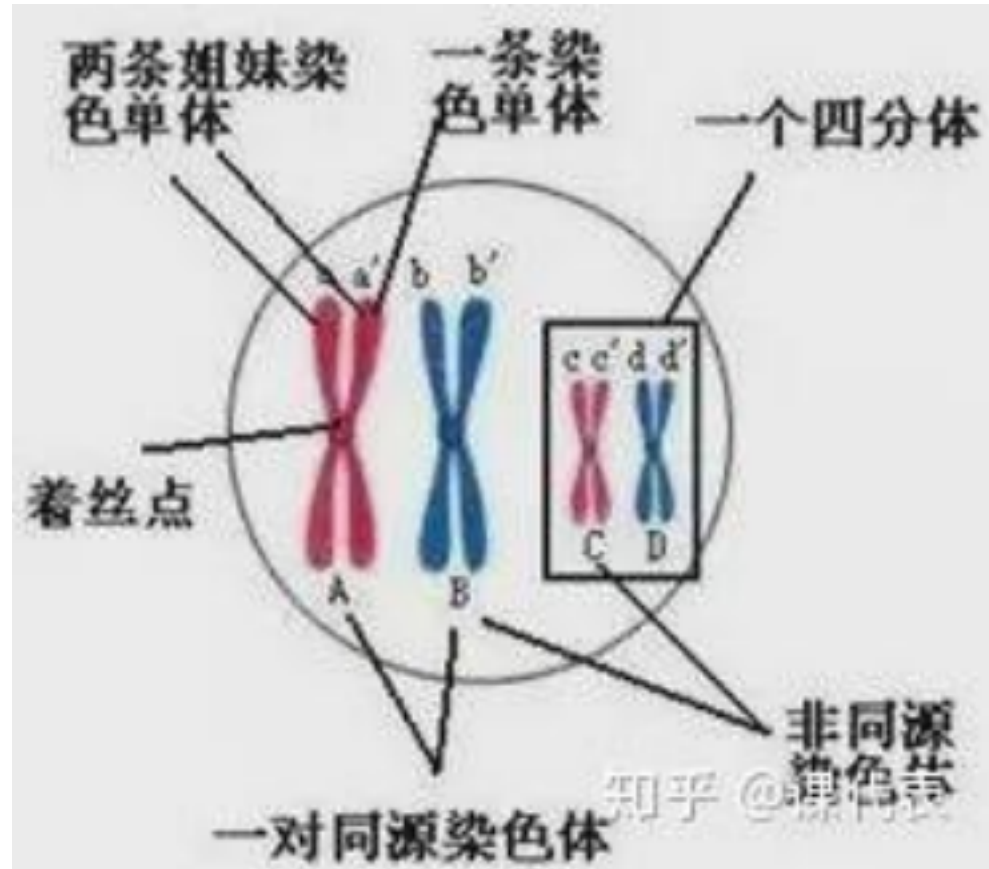
Wall of parent cell



(b) Cell plate formation in a plant cell (TEM)



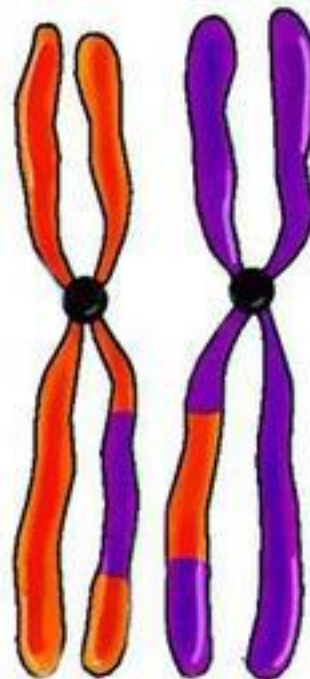
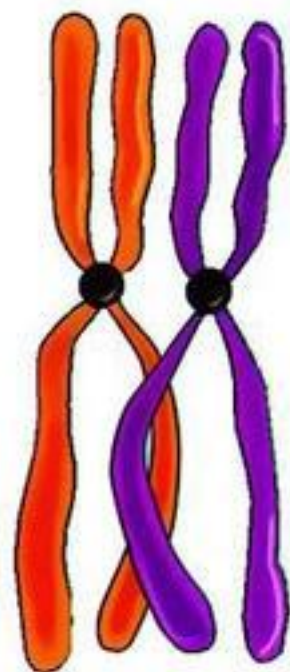
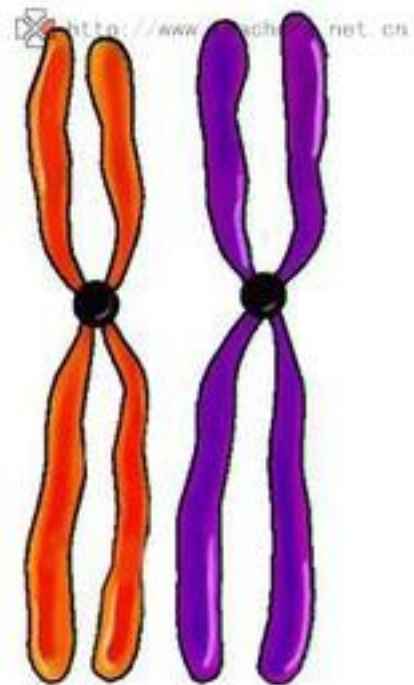
前期 I



减数分裂 I (meiosis I)

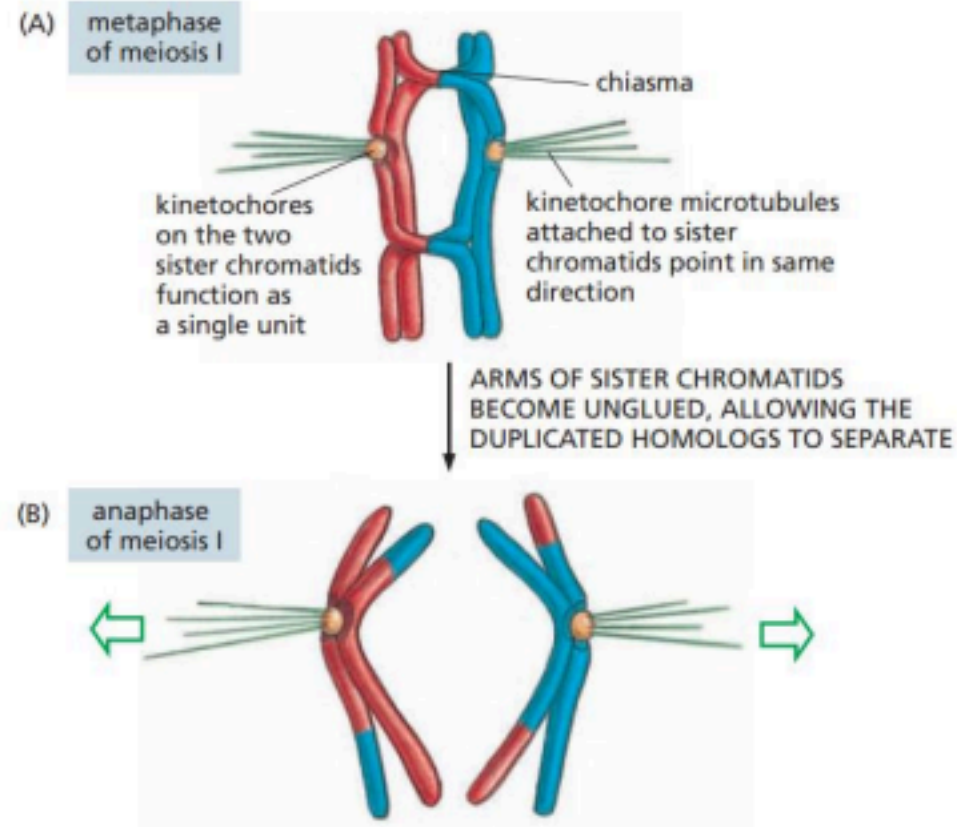
染色体数量=
 染色体数量=
 姐妹染色单体数量=
 姐妹染色单体数量=
 DNA数量=

DNA数量=

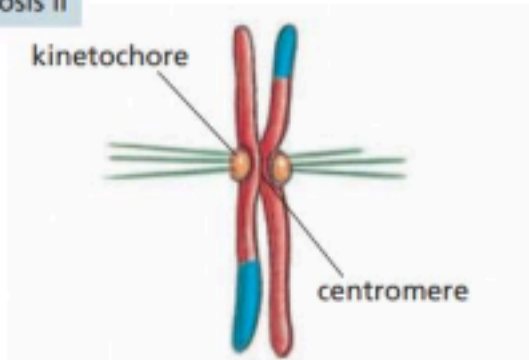


染色体的交叉交换

Figure 19–13 Chiasmata help ensure proper segregation of duplicated homologs during the first meiotic division. (A) In metaphase of meiosis I, chiasmata created by crossing-over hold the maternal and paternal homologs together. At this stage, cohesin proteins keep the sister chromatids glued together along their entire length (see Figure 19–10). The kinetochores of sister chromatids function as a single unit in meiosis I, and microtubules that attach to them point toward the same spindle pole. (B) At anaphase of meiosis I, the cohesins holding the arms of the sister chromatids together are suddenly degraded, allowing the homologs to be separated. Cohesins at the centromere continue to hold the sister chromatids together as the homologs are pulled apart.



(A) metaphase of meiosis II



↓ COHESINS AT CENTROMERE ARE DEGRADED; SISTER CHROMATIDS SEPARATE

(B) anaphase of meiosis II

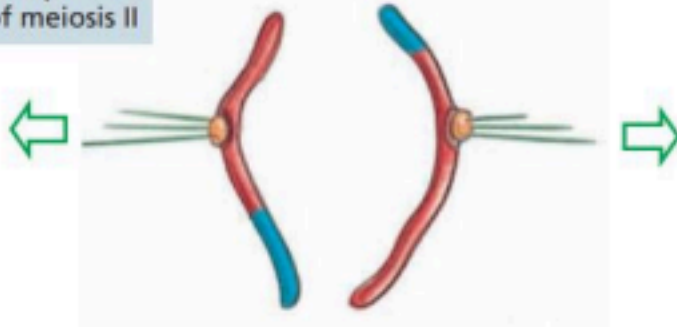
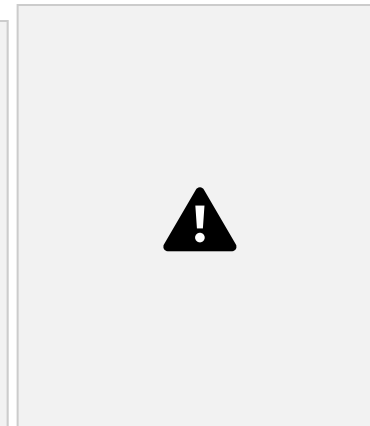
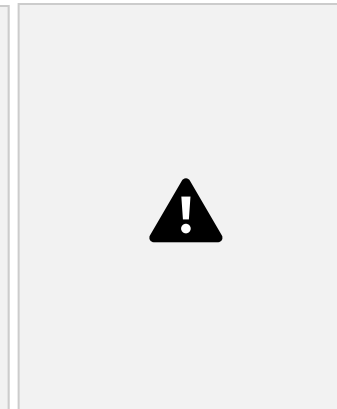
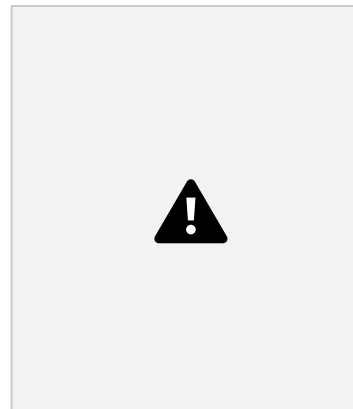
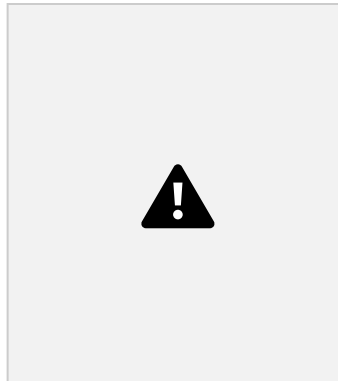


Figure 19–14 In meiosis II, as in mitosis, the kinetochores on each sister chromatid function independently, allowing the two sister chromatids to be pulled to opposite poles. (A) In metaphase of meiosis II, the kinetochores of the sister chromatids point in opposite directions. (B) At anaphase of meiosis II, the cohesins holding the sister chromatids together at the centromere are degraded, allowing kinetochore microtubules to pull the sister chromatids to opposite poles.



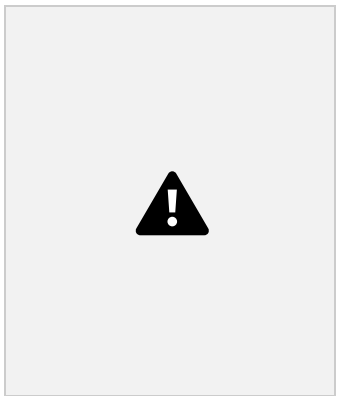


可

能产生那两组子细胞？

- 1. J+M , H+G
 - 2. B+D, C+F
- A,E,K

那些是不可能产生的子细胞？





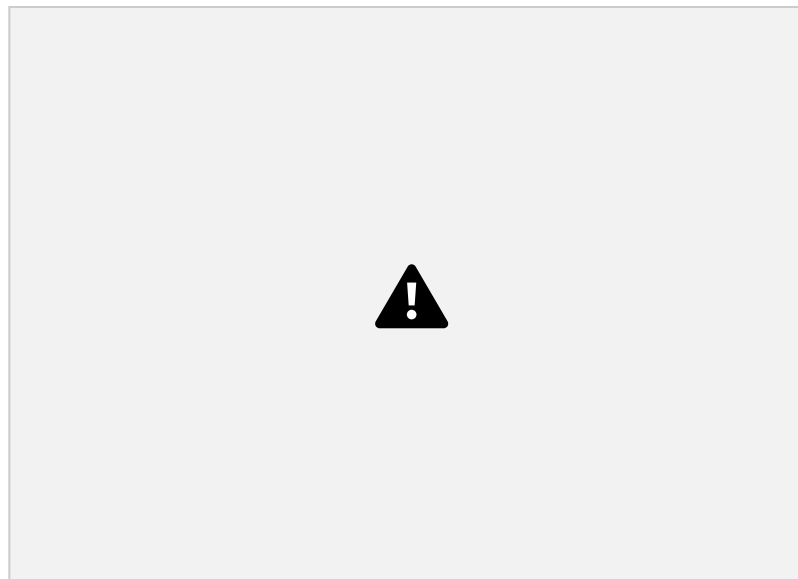
减数分裂(meiosis)

- 1.发生范围 ~有性生殖产生配子的过程
- 2.位置 ~动物的精巢与卵巢、植物的花药与胚珠
- 3.实质~染色体复制一次 ,细胞连续分裂两次
- 4.结果~子细胞的染色体数目是母细胞的一半
→产生单倍的配子



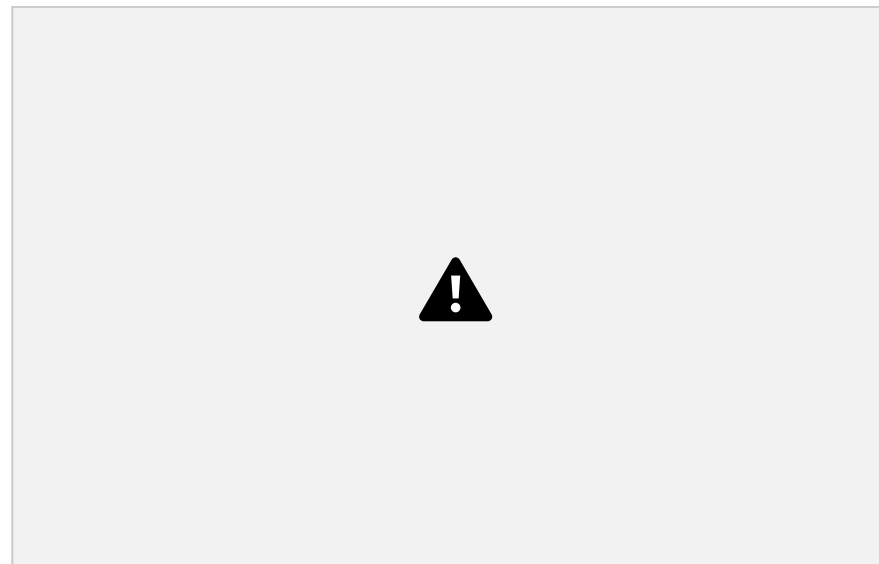


A



减数分裂染色体数目变化

B



减数分裂**DNA**数目变化

C



D 有丝分裂染色体数目变化



有丝分裂**DNA**数目变化





显性突变 隐性突变

原癌基因

抑癌基因



原癌基因



